

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		15132	
Total Threats Mitigated	16	Total Packets Analyzed		1297294664	
High Priority	11	Malicious Packets		1296451988	
Low Priority	5	True Negatives Passed		11420	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	5	UDP Flood		13	
FP Rate	0.04%	Uncertain		3	
IF Precision	99.87%				
IF Recall	96.11%	Action		Count	
IF F1-Score	97.95%	Quarantined		4	
IF Accuracy	98.98%	Time Ban		3	
		Blackhole		0	
		Released		2	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.87%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	96.11%	Share of actual attacks successfully flagged
F1-Score	97.95%	Balanced measure combining Precision and Recall
Accuracy	98.98%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.04%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	3.89%	Actual attacks that went undetected
True Positive Rate (TPR)	96.11%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.96%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3707	FN 150
Actual: Normal	FP 5	TN 11270

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	3707

2c. Response Latency

Metric	Value	Description
Detection Time	360.9 ms	Average time taken to detect an attack
Mitigation Response Time	8.9 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.08%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.64%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.01%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.22	3	3	2026-09-11 11:15:57	2026-09-11 11:17:19
10.0.0.6	3	3	2026-09-11 11:12:43	2026-09-11 11:15:06
10.0.0.20	2	2	2026-09-11 11:15:56	2026-09-11 11:17:07
10.0.0.21	2	2	2026-09-11 11:15:56	2026-09-11 11:17:07

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-11 11:12:43	10.0.0.6	DDoS	UDP Flood	74.1%	High	Time Ban (1m)
2026-09-11 11:13:43	10.0.0.6	DDoS	UDP Flood	74.1%	Medium	Released
2026-09-11 11:14:16	10.0.0.6	DDoS	UDP Flood	74.1%	High	Quarantined
2026-09-11 11:15:56	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 11:15:56	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 11:15:57	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 11:16:06	10.0.0.20	DDoS	UDP Flood	75.8%	High	Time Ban (1m)
2026-09-11 11:16:06	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (1m)
2026-09-11 11:16:07	10.0.0.20	DDoS	UDP Flood	75.8%	High	Time Ban
2026-09-11 11:16:07	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-11 11:16:08	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-11 11:16:08	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-11 11:17:07	10.0.0.20	DDoS	UDP Flood	77.6%	High	Time Ban (2m)
2026-09-11 11:17:07	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (2m)
2026-09-11 11:17:08	10.0.0.22	DDoS	UDP Flood	86.0%	High	Released
2026-09-11 11:17:19	10.0.0.22	DDoS	UDP Flood	86.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6273	86.0%	Critical	Phase 2	3	11s	2026-09-11 11:17:19	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6106	86.0%	High	Phase 2	1	1m 11s	2026-09-11 11:17:08	Ban Expired
10.0.0.20	UDP Flood	0.6373	77.6%	High	Phase 2	2	1m 11s	2026-09-11 11:17:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6220	78.1%	High	Phase 2	2	1m 11s	2026-09-11 11:17:07	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6376	77.7%	High	Phase 2	1	11s	2026-09-11 11:16:08	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6336	75.8%	High	Phase 2	1	10s	2026-09-11 11:16:06	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6288	78.1%	High	Phase 2	1	10s	2026-09-11 11:16:06	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6099	92.0%	High	Phase 1	3	10s	2026-09-11 11:15:06	Attack Stopped
10.0.0.6	UDP Flood	0.6155	74.1%	High	Phase 1	2	10s	2026-09-11 11:14:26	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.6	UDP Flood	0.5576	74.1%	Medium	Phase 2	1	1m 0s	2026-09-11 11:13:43	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE